

Rydberg gas as an RF antenna

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This abstract is abstract.

I. INTRODUCTION

II. THEORY

The four eigenstates of the bare atomic Hamiltonian H_0 involved in the DEIT process are $\{|1\rangle, |2\rangle, |3\rangle, |4\rangle\}$, with associated energies $\hbar\omega_j$, i.e. $H_0|j\rangle = \hbar\omega_j|j\rangle$. Including the interaction provoked by the external electric fields $\mathbf{E}_i$, the full Hamiltonian takes the form

$$\mathcal{H} = H_0 + H_E, \quad H_E \equiv -\hat{\mathbf{d}} \cdot \sum_i \mathbf{E}_i,$$

where $\hat{\mathbf{d}}$ is the electric dipole moment of the Rydberg atom. The atom is exposed to three different electric fields; A weak probe field $\mathbf{E}_p$ is established by the probe laser, an intense dressing field $\mathbf{E}_c$ is generated by the coupling laser, and a separate field $\mathbf{E}_{\text{RF}}$ is created by the RF source.

For simplicity, we treat all fields as plane waves,

$$\mathbf{E}_j = \frac{\mathcal{E}_j}{2} (e^{i\omega_j t} + e^{-i\omega_j t}).$$

The probe laser drives the $|1\rangle \rightarrow |2\rangle$ transition, the coupling laser drives $|2\rangle \rightarrow |3\rangle$, and the RF pumping field drives $|3\rangle \rightarrow |4\rangle$. Accordingly, the field frequencies are near resonance with their respective transitions: $\omega_p \simeq \omega_{21}$, $\omega_c \simeq \omega_{32}$, and $\omega_{\text{RF}} \simeq \omega_{43}$, where we define $\omega_{nm} = \omega_n - \omega_m$ to be the transition frequency from $|m\rangle$ to $|n\rangle$.

We will also need the Rabi frequencies associated with each transition. They are in our case given by

$$\begin{aligned} \Omega_p &= \frac{\langle 2 | \hat{\mathbf{d}} \cdot \mathcal{E}_p | 1 \rangle}{\hbar}, \\ \Omega_c &= \frac{\langle 3 | \hat{\mathbf{d}} \cdot \mathcal{E}_c | 2 \rangle}{\hbar}, \\ \Omega_{\text{RF}} &= \frac{\langle 4 | \hat{\mathbf{d}} \cdot \mathcal{E}_{\text{RF}} | 3 \rangle}{\hbar}. \end{aligned}$$

Working in the interaction picture with respect to the bare Hamiltonian H_0 , a general state of the atom in this setup is given by

$$\begin{aligned} |\psi(t)\rangle &= \sum_{n=1}^4 c_n e^{-iH_0 t/\hbar} |n\rangle = \sum_{n=1}^4 c_n e^{-i\omega_n t} |n\rangle, \\ \partial_t |\psi(t)\rangle &= \sum_{n=1}^4 (\dot{c}_n - i\omega_n c_n) e^{-i\omega_n t} |n\rangle. \end{aligned}$$

The Schrödinger equation then reads $i\hbar\partial_t |\psi\rangle = \hat{H}_E |\psi\rangle$, where $\hat{H}_E$ is the interaction Hamiltonian in the interaction picture, i.e.

$$\hat{H}_E(t) = e^{iH_0 t/\hbar} H_E(t) e^{-iH_0 t/\hbar}.$$

The matrix elements of the interaction picture Hamiltonian in the $\{|1\rangle, |2\rangle, |3\rangle, |4\rangle\}$ basis are given by

$$(\hat{H}_E)_{nm} = \langle n | \hat{H}_E | m \rangle = e^{i\omega_{nm} t} \langle n | H_E | m \rangle.$$

The non-zero elements are those associated with the three driven transitions. As an example we can calculate the (1,2)-element:

$$\begin{aligned} (\hat{H}_E)_{12} &= e^{i\omega_{12} t} \langle 1 | H_E | 2 \rangle \\ &= \frac{1}{2} e^{-i\omega_{21} t} \langle 1 | \hat{\mathbf{d}} | 2 \rangle \cdot \mathcal{E}_p (e^{i\omega_p t} + e^{-i\omega_p t}) \\ &= \frac{1}{2} \hbar \Omega_p (e^{-i(\omega_{21} - \omega_p) t} + e^{-i(\omega_{21} + \omega_p) t}). \end{aligned}$$

The rapidly oscillating $e^{-i(\omega_{21} + \omega_p) t}$ term will be washed out and averaged to zero over any reasonable measurement period and can thus be neglected. Rewriting the surviving term using the detuning parameter $\Delta_p = \omega_{21} - \omega_p$, we find

$$(\hat{H}_E)_{12} = (\hat{H}_E)_{12}^\dagger = \frac{1}{2} \hbar \Omega_p e^{-i\Delta_p t}.$$

The other matrix elements can be found completely analogously and in the end we get

$$\hat{H}_E = \frac{\hbar}{2} \begin{bmatrix} 0 & \Omega_p e^{-i\Delta_p t} & 0 & 0 \\ \Omega_p e^{i\Delta_p t} & 0 & \Omega_c e^{-i\Delta_c t} & 0 \\ 0 & \Omega_c e^{i\Delta_c t} & 0 & \Omega_{\text{RF}} e^{-i\Delta_{\text{RF}} t} \\ 0 & 0 & \Omega_{\text{RF}} e^{i\Delta_{\text{RF}} t} & 0 \end{bmatrix}.$$

We can get rid of the time-dependence of the Hamiltonian by transforming to a rotating reference frame using the unitary transformation matrix

$$U \equiv \text{diag}(1, e^{-i\Delta_p t}, e^{-i(\Delta_p + \Delta_c) t}, e^{-i(\Delta_p + \Delta_c + \Delta_{\text{RF}}) t}).$$

The resulting Hamiltonian in the rotating frame can be determined to be

$$\begin{aligned} \tilde{H}_E &= U^\dagger \hat{H}_E U - i\hbar U^\dagger \dot{U} \\ &= \hbar \begin{bmatrix} 0 & \Omega_p/2 & 0 & 0 \\ \Omega_p/2 & \Delta_p & \Omega_c/2 & 0 \\ 0 & \Omega_c/2 & \Delta_p + \Delta_c & \Omega_{\text{RF}}/2 \\ 0 & 0 & \Omega_{\text{RF}}/2 & \Delta_p + \Delta_c + \Delta_{\text{RF}} \end{bmatrix}. \quad (1) \end{aligned}$$

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A. Production of two electromagnetically induced transparency (EIT) windows

Under the assumption of a weak probe laser ($|\mathbf{E}_p| \ll 1$), the Rabi frequency $\Omega_p \propto |\mathbf{E}_p|$ is negligible. For simplicity we can also assume perfectly on-resonance coupling and RF pumping, i.e. $\Delta_c = \Delta_{\text{RF}} = 0$. Let us only consider the effective Hamiltonian which in this case will be the lower 3×3 -submatrix of Eq.(1),

$$H_{\text{eff}} = \hbar \begin{bmatrix} \Delta_p & \Omega_c/2 & 0 \\ \Omega_c/2 & \Delta_p & \Omega_{\text{RF}}/2 \\ 0 & \Omega_{\text{RF}}/2 & \Delta_p \end{bmatrix}.$$

Since the Δ_p 's along the diagonal only will uniformly shift the energy levels, we can neglect them when calculating the eigenstate energies. Doing this, we get three eigenstates with eigenvalues

$$\lambda_0 = 0, \quad \lambda_{\pm} = \pm \frac{\Omega_{\text{eff}}}{2},$$

with $\Omega_{\text{eff}} \equiv \sqrt{\Omega_{\text{RF}}^2 + \Omega_c^2}$. The eigenstate associated with λ_0 is the so-called *dark state* and can be represented as

$$|\Psi_0\rangle = -\frac{\Omega_{\text{RF}}}{\Omega_{\text{eff}}} |2\rangle + \frac{\Omega_c}{\Omega_{\text{eff}}} |4\rangle.$$

The two other eigenstates, the *bright states*, can be represented as

$$|\Psi_{\pm}\rangle = \frac{\Omega_c}{\sqrt{2}\Omega_{\text{eff}}} |2\rangle \pm \frac{1}{\sqrt{2}} |3\rangle + \frac{\Omega_{\text{RF}}}{\sqrt{2}\Omega_{\text{eff}}} |4\rangle.$$

The transition amplitudes of the probe to each of the dressed states is now given by the coupling constants and

can be found as the projection of the $|2\rangle$ state onto each of the eigenstates,

$$c_0 = \langle 2|\Psi_0\rangle = -\frac{\Omega_{\text{RF}}}{\Omega_{\text{eff}}},$$

$$c_{\pm} = \langle 2|\Psi_{\pm}\rangle = \frac{\Omega_c}{\sqrt{2}\Omega_{\text{eff}}}.$$

The susceptibility is shown to be

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$$\chi \propto \sum_{i=0,-,+} \frac{|c_i|^2}{\Delta_p - \lambda_i},$$

which in our case can be expanded and simplified to

$$\chi \propto \frac{\Delta_p^2 - \Omega_{\text{RF}}^2/4}{\Delta_p(\Delta_p^2 - \Omega_{\text{eff}}^2/4)}.$$

The absorption peaks when the denominator goes to zero. We therefore have three such peaks, corresponding to $\Delta_p = 0$ and $\Delta_p = \pm\Omega_{\text{eff}}/2$. Crucially, we also get two transparency windows around $\chi = 0$ which are centered at $\Delta_p = \pm\Omega_{\text{RF}}/2$. These are the two EIT windows.

III. METHOD

IV. RESULTS

V. DISCUSSION

VI. CONCLUSION

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